

Power BI Basics

Question 1: What is Power BI and why is it used in businesses?

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps users connect to different data sources, transform data, and create interactive dashboards and reports.

Businesses use Power BI to:

- Analyze large amounts of data easily.
- Create visual dashboards and reports for decision making.
- Track business performance through KPIs.
- Share insights across teams and departments.
- Connect data from multiple sources such as Excel, databases, and cloud services.

Overall, Power BI helps organizations make data-driven decisions quickly and efficiently.

Question 2: Name and explain the three main components of Power BI.

The three main components of Power BI are:

1. Power BI Desktop

Power BI Desktop is a Windows application used to create reports and dashboards. It allows users to connect to data sources, clean and transform data, and build visualizations.

2. Power BI Service

Power BI Service is the cloud-based platform where reports and dashboards are published and shared with others. Users can collaborate, schedule refreshes, and access reports through a web browser.

3. Power BI Mobile

Power BI Mobile allows users to view dashboards and reports on mobile devices such as smartphones and tablets. It helps users monitor business data anytime and anywhere.

Question 3: Explain the Power BI workflow.

The Power BI workflow consists of several steps:

1. Data Collection – Data is gathered from various sources such as Excel files, databases, APIs, or cloud services.
2. Data Transformation – The data is cleaned and transformed using Power Query to remove errors, duplicates, and unnecessary columns.
3. Data Modeling – Relationships between tables are created and calculations are defined using DAX formulas.
4. Data Visualization – Interactive charts, tables, and dashboards are created in the Report View.
5. Publishing – The report is published to the Power BI Service.
6. Sharing – Reports and dashboards are shared with other users or teams in the organization.

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Four common data cleaning tasks in Power Query are:

1. Removing duplicate rows.
2. Replacing or handling missing values.
3. Changing data types (text, number, date, etc.).
4. Splitting or merging columns.
5. Removing unnecessary columns or rows.
6. Renaming columns for better understanding.

These tasks help prepare the data for accurate analysis and reporting.

Question 5: Write step by step instructions to load the above dataset.

Steps to load a dataset into Power BI:

1. Open Power BI Desktop.
2. Click on "Get Data" from the Home tab.
3. Choose the data source type (for example Excel, CSV, or database).
4. Browse and select the dataset file (Real_Estate dataset).
5. Click "Open".
6. A preview window will appear showing the dataset.
7. Click "Load" to import the data into Power BI.
8. The dataset will now appear in the Fields pane and can be used to create reports and visualizations.

Question 6: Define Data View, Report View, and Model View. Explain the purpose of each view.

1. Data View

Data View allows users to see the data in a table format inside Power BI. It is mainly used to inspect the data, create calculated columns, and verify values.

2. Report View

Report View is used to create visualizations such as charts, graphs, tables, and dashboards. Users design interactive reports in this view.

3. Model View

Model View shows the relationships between different tables in the dataset. It helps users manage data models and create relationships between tables for better analysis.

Question 7: Discuss the different data sources that Power BI supports.

Power BI supports a wide variety of data sources including:

1. Files – Excel, CSV, XML, JSON, and Text files.
2. Databases – SQL Server, MySQL, Oracle, PostgreSQL, and others.
3. Cloud Services – Google Analytics, Azure, SharePoint, Salesforce, etc.
4. Online Services – Web APIs and online data feeds.
5. Other Sources – Power BI datasets, OData feeds, and data warehouses.

This flexibility allows businesses to integrate data from multiple systems into a single dashboard.

Question 8: Split Owner Name to create two new columns as First Name and Last Name.

Steps to split Owner Name in Power BI:

1. Open Power BI Desktop.
2. Go to the Home tab and click "Transform Data" to open Power Query Editor.
3. Select the column named "Owner Name".
4. Go to the Transform tab.
5. Click "Split Column".
6. Choose "By Delimiter".
7. Select the delimiter as Space.
8. Power BI will create two columns.
9. Rename the first column as "First Name".
10. Rename the second column as "Last Name".
11. Click "Close & Apply" to apply the changes.

Now the Owner Name column is split into First Name and Last Name columns.